REVIEW ARTICLE

Clear or almost clear skin improves the quality of life in patients with moderate-to-severe psoriasis: a systematic review and meta-analysis

L. Puig,[1,*] H. Thom,[2] P. Mollon,[3] H. Tian,[4] G.S. Ramakrishna[5]

[1]Department of Dermatology, Hospital de la Santa Creu i Sant Pau, Universitat Autonoma de Barcelona Medical School, Barcelona, Spain
[2]School of Social and Community Medicine, University of Bristol, Bristol, UK
[3]Global Patient Access, Novartis Pharma AG, Basel, Switzerland
[4]Global Medical Affairs, Novartis Pharmaceuticals Corporation, One Health Plaza, East Hanover, NJ,USA
[5]Patient Access Services, Novartis Healthcare Private Limited, Hyderabad, India
[*]**Correspondence:** L. Puig. E-mail: LPuig@santpau.cat

Abstract

Psoriasis Area and Severity Index (PASI) 75 response is currently considered the gold standard for assessing treatment efficacy in moderate-to-severe psoriatic patients. PASI 90 response denotes better clinical improvement compared to PASI 75. Very few studies have assessed if a greater PASI clinical response is associated with greater improvements in Dermatology Life Quality Index (DLQI). A systematic review and meta-analysis was performed to assess the association between PASI response and DLQI. The study was conducted to assess whether greater improvement in PASI scores from PASI 75–89 to PASI 90 is associated with greater Quality of life (QoL) improvements, specifically DLQI scores. Systematic searches were conducted in MEDLINE, EMBASE and Cochrane Library to identify studies evaluating biologic interventions in adult moderate-to-severe psoriasis patients reporting PASI response and their corresponding DLQI change from baseline score. The quality of evidence was assessed through Jadad score for randomized controlled trials and Downs and Black's checklist for observational studies. Meta-analysis estimated change from baseline in DLQI for PASI 75–89 responders to be 78% (95% credible intervals [CrI]: 75–82%) and for PASI 90 responders to be 90% (95% CrI: 88–93%). This implies 12% greater improvement in DLQI score for PASI 90 responders compared with PASI 75–89 responders. In addition, the meta-analysis also showed a statistically significant difference in DLQI score of 0/1 between PASI 75-89 and PASI 90 responders (45% [95% CrI]; 41.0–50.0% and 73% [95% CrI]; 70.0–76.0%), respectively, Bayesian $P < 0.0001$). In conclusion, substantial improvement in clinical efficacy is associated with improved QoL in patients with moderate-to-severe psoriasis suggesting that PASI 90 responders (clear or almost clear skin) could achieve a superior QoL compared to PASI 75–89 responders.

Received: 25 May 2016; Accepted: 22 July 2016

Conflicts of interest

This study was sponsored by Novartis Pharma AG, Basel, Switzerland. Luis Puig- consultancy honoraria and participation in clinical trials sponsored by Novartis. Howard Thom- consultant for Novartis Pharma AG and Eli-Lilly; Patrick Mollon and Haijun Tian are employees of Novartis Pharma AG, and Ramakrishna GS is employee of Novartis Healthcare Private Limited

Funding sources

This study was sponsored by Novartis Pharma AG, Basel, Switzerland.

Introduction

Psoriasis, a chronic, relapsing, immune-mediated, inflammatory skin disease affecting approximately 2–3% of the population is associated with a significant disease burden.[1] A report by the International Federation of Psoriasis Associations indicates that in the US there is an approximate loss of 56 million hours of work and USD 2–3 billions spent on treatment every year (http://www.ifpa-pso.org/, accessed 18 Jan 2016). The World Health Organization (WHO) also recognizes psoriasis as a serious non-communicable disease. Patients with psoriasis

 © 2016 European Academy of Dermatology and Venereology

also have an increased risk of developing comorbid conditions such as psoriatic arthritis, cardiovascular diseases, infections, malignancies and autoimmune diseases.[2] Therefore, patients with psoriasis have a significantly compromised health-related quality of life (QoL),[3] leaving them stressed, socially withdrawn, and importantly, with suicidal tendency.[4–7] Approximately 25–44% of patients have moderate-to-severe disease,[8] and need an aggressive treatment due to increased severity of symptoms.[9,10]

The goal of treatment is to reduce the signs and symptoms of the disease, possibly slowing the natural progression of the disease. Available systemic biologic treatments target the pathophysiological pathways modulated by several cytokines.[11] Currently available options available in this class of drugs include anti-TNF agents (infliximab, adalimumab and etanercept), anti-interleukin (IL)-12/23 (ustekinumab) and anti-IL-17A agents (secukinumab).[12,13] The Psoriasis Area and Severity Index (PASI) remains the most common efficacy endpoint in clinical trials and is considered as a gold standard since it accounts for both severity (erythema, induration and desquamation) and extent of disease (head, trunk, arms and legs).[14] However, the benefit remains incomplete if this does not translate into improved quality of life. Therefore, evaluating the impact on patients' QoL following treatment, and therefore, any reduction in the disease burden seems one of the most important aspects of treatment considering the tormenting impact of psoriasis. This is especially pertinent since treatment with biologic agents is costly, and would be appreciated when reflects in an improvement in QoL.

Keeping this in view, we performed a systematic review and meta-analysis to understand if the improvement in PASI scores results in an overall improved QoL in adult patients with moderate-to-severe psoriasis where patients were treated with one of the approved systemic biologic agents. PASI was specifically chosen as the clinical efficacy endpoint of interest as it is the most commonly used score in psoriasis trials, making meta-analysis of existing trials possible.[15] A 75% or greater improvement from baseline (PASI 75) is widely accepted as a useful and realistic clinical outcome. Importantly, PASI 90, meaning a 90% or greater improvement from baseline, fulfils the European regulatory definition of therapeutic success[16] and appears to be better correlated with improved health-related QoL compared with PASI 75.[17] For evaluating the impact on QoL, we used Dermatology Life Quality Index (DLQI),[18] a commonly used PRO in psoriasis trials.[15] The treatment goal for DLQI can be defined as a score of 0 or 1, implying that patient's QoL is not affected by psoriasis.[19] Patients achieving PASI 90 will have better clinical improvement compared to PASI 75. Almost clear skin (PASI 90) will be beneficial for patients if it leads to better quality of life (DLQI score). Here, we present the results from this systematic review, and meta-analysis from five clinical trials.

Materials and Methods

This systematic review and meta-analysis were conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) statement.[20] The selection of databases, eligibility criteria, outcomes of the review and analyses methods were defined *a priori* in an internal protocol. Study selection, data extraction and quality assessment were performed independently by two assessors. Any discrepancy was reconciled by a third independent reviewer.

Searches and data extraction

A comprehensive literature search of MEDLINE (including Medline-in-Process), PubMed, EMBASE, the Cochrane Centre Register of Controlled Trials and National Health Service Economic Evaluation Database was carried out in Ovid for relevant articles for last 10 years (2004–2014). The detailed search strategy is presented in Table 1.

Published articles were included if they investigated one of the following treatments: all disease-modifying anti-rheumatic drugs (DMARDs) including biologics used for psoriasis treatment. The comparisons had to be done either vs. a placebo or an active treatment. Study design included randomized clinical trials (RCTs) with any blinding status, non-RCTs and observational studies. There was no restriction on treatment duration. Patient population consisted of adults (≥18 years of age) of any gender/race with moderate-to-severe psoriasis. The inclusion criteria in the clinical trials and studies were in line with the definition of moderate-to-severe psoriasis. Studies had to report at least one outcome of interest, that is, PASI 75–89 (≥75–<90% improvement (reduction) in PASI score with respect to baseline), PASI 90 (≥90% improvement (reduction) in PASI score), PASI 100 (complete clearance of psoriasis corresponds to absolute score PASI = 0) and their corresponding DLQI 0/1 response rates. Data were segregated by the time-point according to different

Table 1 List of search terms used for retrieving relevant studies from various databases

Search criteria	Search terms/strings
Disease	Exp *psoriasis/
PASI	((psoriasis area and severity index) or (psoriasis area severity index or PASI or PASI score or psoriasis area severity index 75 or PASI 75 or psoriasis area severity index 90 or PASI 90 or psoriasis area severity index 100 or PASI 100)).mp.
Comorbidities	(comorbidit* or co-morbidit*).mp.
Quality of life	(quality of life or quality-of-life or QoL or disab* or morbid* or mortalit* or patient reported outcome or patient-reported outcome or PRO or Quality adjusted life years or QALY or Disability adjusted life years or Quality-adjusted life years or Disability-adjusted life years or DALY or prefer* or questionnaire or scale or score or satisfact* or utilit* or dermatology life quality index or DLQI or EQ-5D).mp.

© 2016 European Academy of Dermatology and Venereology

14683083, 2017, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jdv.14007 by Western University, Wiley Online Library on [12/06/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

outcomes reported, both for the randomized phase or the long-term maintenance phase.

Only articles in English language were included. Exclusion criteria were as follows: use of conventional interventions, no subgroup analysis for disease of interest and studies not reporting secondary outcomes data (QoL) for the primary outcome of interest (PASI response).

After exclusion of duplicates, titles and abstracts were screened for inclusion and exclusion. Potentially relevant articles were checked in full text for inclusion. Study characteristics (e.g. study design, phase, blinding status, sample size), study population (e.g. age, sex, duration of psoriasis), and study results of included trials were extracted using a standardized form. Primary outcomes were PASI response defined as PASI 75–89, PASI 90 and PASI 100. Secondary outcomes included DLQI response of 0/1, defined as final DLQI equal to 0 or 1, DLQI score mean percentage change from baseline and DLQI score mean change from baseline.

Quality assessment

The Jadad scoring system was used to appraise adequacy of randomization, blinding and reporting of withdrawals/discontinuations for RCTs, while the Downs and Black's checklist was used to appraise the non-RCTs. Four studies attained a maximum score of 5 on the Jadad scale, suggesting adequate reporting of randomization, blinding and reporting of withdrawals/discontinuations. While one study[17] had a score of 3, no other study reported a Jadad score of less than 4, indicating that studies included in the review were of good quality. The use of adequate method of allocation concealment was reported in all the included studies, resulting in allocation concealment grade 'A'. None of the studies reported an inadequate method of allocation concealment or that method of concealment was not used in the study. The open-label single arm trials were of good quality in terms of reporting and assessment of bias – one study[21] scored 16 and other study[17]scored 18 on the Downs and Black's checklist.

Data analysis

Meta-analysis is a statistical technique to combine multiple estimates of a quantity into a single estimate.[22] It is most commonly used to combine estimates of treatment effects from multiple trials but can be used to synthesize estimates of any quantity, as is often necessary in medical decision-making.[23] For the current study, Bayesian meta-analysis was used to combine estimates of DLQI improvement in PASI responder subgroups when three or more studies are available. If sufficient data were available to fit both, a choice between fixed and random effects models was made on the basis of lowest Deviance Information Criterion (DIC) and residual deviance.[24,25]

The continuous data included DLQI score mean percentage change from baseline and DLQI score mean change from baseline. The dichotomous data were DLQI response of 0/1. Relative risks with 95% CrI for dichotomous data and mean differences with 95% CrI for continuous data were calculated. Classical statistically significant differences between DLQI effect in PASI 75–89 and PASI 90 responders, at 0.05 level, were assessed using one-sided Bayesian *P*-values; these are the Bayesian probability that DLQI effect was greater in one responder group than the other. Meta-analysis of studies was conducted using OpenBUGS (version 3.2.2 rev 1063).[26]

Meta-analysis of studies reporting DLQI percentage change from baseline and DLQI response of 0/1 was conducted for PASI 75-89 and PASI 90 separately. Sample size-weighted averages were used to impute mean and standard deviation of DLQI percentage change from baseline in studies that did not directly report for PASI 75-89 or PASI 90.

Results

Systematic search in databases (MEDLINE, EMBASE and Cochrane Library) yielded 1085 results. Due to the overlap of records across the databases searched, 292 abstracts were identified as duplicates. Following the first screening of these citations according to their titles and abstracts, 264 potentially relevant references were identified. A detailed evaluation of full-text reports of these citations was performed, and 251 studies were excluded due to non-reporting of the secondary outcome data (i.e. DLQI) for the patients achieving PASI 75–89 and PASI 90 responses. None of the studies reported secondary outcome data for PASI 100 (Fig. 1). Of the 13 studies extracted, four studies met the predefined eligibility criteria for inclusion in this meta-analysis, and all included patients with moderate–to-severe psoriasis. Apart from the four studies identified from systematic literature search, recently published results from CLEAR study were also included in the meta-analysis. CLEAR, a head-to-head trial, compared the efficacy and safety of secukinumab with ustekinumab in subjects with moderate-to-severe plaque psoriasis.

Primary objective of CLEAR study was to demonstrate superiority of secukinumab vs. ustekinumab with respect to PASI 90 response at week 16. DLQI response was also reported in the study.[27] The details of the demographics and disease characteristics of study participants at baseline are listed in Table 2. Baseline demographics and clinical disease characteristics were comparable across the included studies. The mean age of the patients in the RCTs was about 40 years, and >65% patients were male. The mean duration of disease ranged from 15 to 18.8 years.

Qualitative analysis

Patients achieving primary efficacy endpoint (10–16 weeks)

Studies included in the analysis of the randomized phase of the

© 2016 European Academy of Dermatology and Venereology

14683083, 2017, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jdv.14007 by Western University, Wiley Online Library on [12/06/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

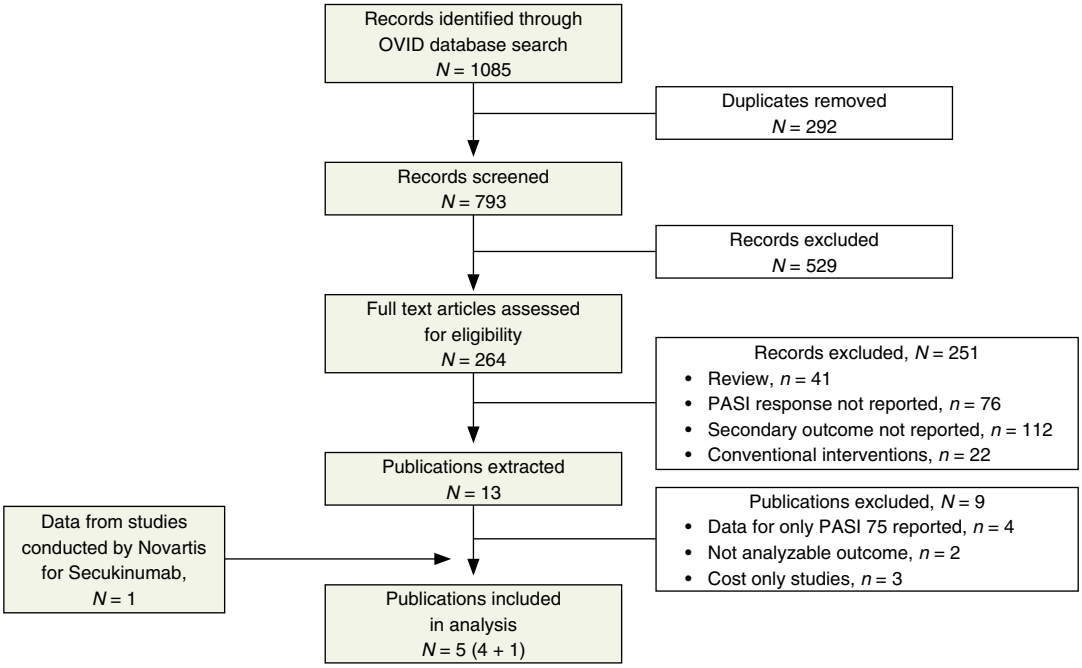


Figure 1 PRISMA Flow for study inclusion. PRISMA, preferred reporting items for systematic reviews and meta-analyses.

Table 2 The demographics and disease characteristics of study participants at baseline

Trial Name	Interventions	Number of participants	Gender		Age (years) Mean (SD)	Duration of psoriasis (years) Mean (SD)
			Male n (%)	Female n (%)		
PSUNRISE trial CHAMPION & REVEAL trials CHAMPION trial	Infliximab 5 mg/kg	215	137 (63.7)		44.4 (13.3)	18.8 (11.4)
	Adalimumab pooled	1469	971 (66.1)	498 (33.9)	44.1 (13.1)	18.6 (11.6)
	Adalimumab 40 mg EOW	108	70 (64.8)	37 (35.2)	44.1 (13.2)	18.6 (12)
	Placebo	53	35 (66)	18 (34.0)	45.4 (13.4)	18.8 (12)
ERASURE trial	Secukinumab 300 mg	245	169 (69.0)	76 (31.0)	44.9 (13.5)	17.4 (11.1)
	Secukinumab 150 mg	245	168 (68.6)	77 (31.4)	44.9 (13.3)	17.5 (12.0)
	Placebo	248	172 (69.4)	76 (30.6)	45.4 (12.6)	17.3 (12.4)
FIXTURE trial	Secukinumab 300 mg	327	224 (68.5)	103 (31.5)	44.5 (13.2)	15.8 (12.3)
	Secukinumab 150 mg	327	236 (72.2)	91 (27.8)	45.4 (12.9)	17.3 (12.2)
	Etanercept 50 mg	326	232 (71.2)	94 (28.8)	43.8 (13.0)	16.4 (12.0)
	Placebo	326	237 (72.7)	89 (27.3)	44.1 (12.6)	16.6 (11.6)
CLEAR trial	Secukinumab 300 mg Ustekinumab	337 339	229 (68) 252 (74.3)	108 (32) 87 (25.7)	45.2 (13.96) 44.6 (13.67)	19.6 (12.9) 16.1 (11.24)

EOW, every other week.

© 2016 European Academy of Dermatology and Venereology

14683083, 2017, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jdv.14007 by Western University, Wiley Online Library on [12/06/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 3 List of studies included in the analysis of the primary efficacy endpoint at 10–16 weeks and meta-analysis

Clinical trials included	Source	Interventions assessed
PSUNRISE (single arm)	Kalb 2013	Infliximab
CHAMPION	Navarini 2014	Adalimumab
REVEAL	Revicki 2007	Adalimumab
ERASURE	Internal	Secukinumab
FIXTURE	Internal	Secukinumab
CLEAR	Internal	Secukinumab

trials assessing the correlation between PASI response and QoL are listed in Table 3.

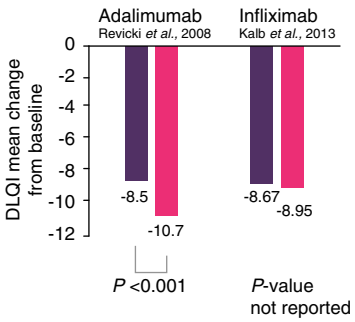
DLQI mean change from baseline

Two studies reported data for mean change from baseline in DLQI score for both PASI 75–89 and PASI 90 responders.[21,28] Kalb *et al.*[21] 2013 only reported DLQI score for PASI ≥ 75 and PASI 90 but PASI 75-89 could be imputed as score in PASI ≥ 75 is the weighted average of PASI 75-89 and PASI 90; the imputed mean DLQI change from baseline was −8.33, with a standard deviation of 14.46 and sample size of 50.[21] Score for PASI 90 in the study by Revicki *et al.*[28] was imputed as the weighted average of PASI 90–99 and PASI ≥ 100; the mean change was −10.48 with a standard deviation of 5.11 and sample size of 447.

Meta-analysing these results, the mean reduction from baseline in DLQI score, that is, improvement, was greater for PASI 90 responders as compared with PASI 75–89 responders (Fig. 2). This difference was statistically significant in the pooled CHAMPION and REVEAL trials.[28]

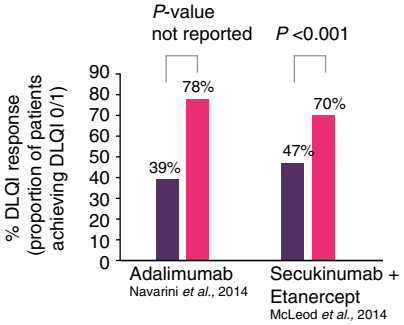
DLQI percent improvement from baseline

Only one study reported data for mean percent improvement from baseline in DLQI score for both PASI 75–89 and PASI 90 responders. The mean percent improvement from baseline in DLQI score was greater for PASI 90 responders as compared with PASI 75–89 responders (91% vs. 75% respectively).[29]



PASI 75-89 responders	-8.5	-8.67
PASI 90 responders	-10.7	-8.95

Figure 2 Mean change in DLQI score from baseline by PASI responders. DLQI, dermatology life quality index.



PASI 75-89 responders	39%	47%
PASI 90 responders	78%	70%

Figure 3 DLQI response by patients achieving PASI response. DLQI, dermatology life quality index; PASI, psoriasis area and severity index.

DLQI 0/1 response

Two studies reported data for DLQI response of 0/1 for PASI 75–89 and PASI 90 responders.[29,30] In both studies, greater proportion of PASI 90 responders achieved a DLQI response of 0/1 as compared with PASI 75–89 responders (Fig. 3). This difference was statistically significant for studies with secukinumab ($P < 0.001$).[30]

Patients on maintenance phase (26–52 weeks) Studies included in the analysis of the maintenance phase of the trials assessing the correlation between PASI response and QoL are listed in Table 3.

DLQI 0/1 response

Two studies reported data for DLQI response of 0/1 for PASI 75–89 and PASI 90 responders.[17,30] In both studies, greater proportion of PASI 90 responders achieved a DLQI response of 0/1 as compared with PASI 75–89 responders (74% vs. 56.7% and 85.7% vs. 50% respectively). This difference was statistically significant for studies with secukinumab ($P < 0.001$).[30]

DLQI score mean change from baseline

The mean reduction from baseline in DLQI score, that is, improvement, was greater for PASI 90 responders as compared with PASI 75–89 responders (−8.62 vs. −8.2 respectively) at Week 26.[21]

Meta-analysis Only two studies reported DLQI 0/1 response and one study reported DLQI mean change at maintenance

© 2016 European Academy of Dermatology and Venereology

14683083, 2017, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jdv.14007 by Western University, Wiley Online Library on [12/06/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

phase; so, only the primary efficacy endpoint (10–16 weeks) was meta-analysed.

Patients achieving primary efficacy endpoint (10–16 weeks)

Percentage change in mean DLQI from baseline Four studies including the Novartis CLEAR study reported change in DLQI for PASI 75–89 and PASI 90 responders.[21,28,30] Using a fixed effects model, the mean percentage reduction from baseline in DLQI scores, i.e. improvement, was greater (Bayesian probability >99.99%) in PASI 90 responders (90% [95% Credible Intervals, CrI: 88%, 93%]) as compared with PASI 75–89 responders (78% [95% CrI: 75%, 82%]). A forest plot for the studies included in this meta-analysis is presented in Figure 4.

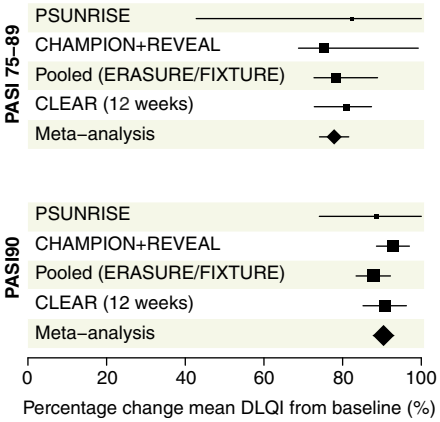


Figure 4 Forest plot: DLQI change from baseline – PASI 90 vs. PASI 75–89. DLQI, dermatology life quality index; PASI, psoriasis area and severity index.

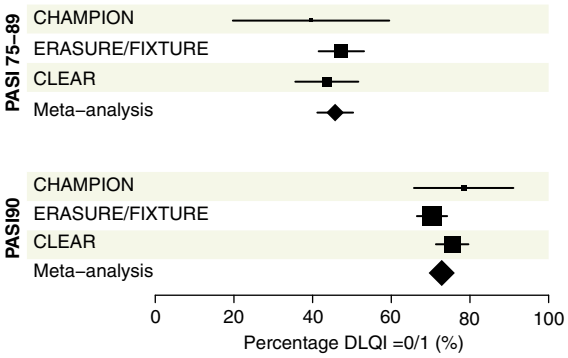


Figure 5 Forest plot: DLQI 0/1 response – PASI 90 vs. PASI 75-89.: DLQI, dermatology life quality index; PASI, psoriasis area and severity index.

DLQI 0/1 response

Three studies including Novartis CLEAR study reported data for DLQI response of 0/1 for PASI 75–89 and PASI 90 responders.[27,29,30] Using a fixed effects model, a greater proportion of PASI 90 responders (Bayesian probability >99.99%) achieved a DLQI score of 0/1 (76% [95% CrI: 70%, 76%]) as compared with PASI 75–89 responders (45% [95% CrI: 41%, 50%]). A forest plot for the studies included in this meta-analysis is presented in Figure 5.

Discussion

The present analysis demonstrated that improvement in QoL as evaluated by DLQI score was commensurate with a higher clinical response as evaluated by PASI response. Overall, the improvement in QoL was greater in PASI 90 responders as compared with PASI 75–89 responders in both short- and long-term. The improvement in QoL was sustained over 26–52 weeks in PASI 90 responders while being in maintenance phase.

The present systematic review and meta-analysis focused on an important outcome in psoriasis trials, that is, the clinical improvement, and the benefit based on PRO, both critical to establishing the therapeutic effectiveness and satisfaction with treatment, especially biologics. This is principally pertinent because of two reasons. First, psoriasis is associated with significant negative impact on emotional, physical, psychological and social functioning, and second is the cost consideration for treatment with biologics.

The data from studies included in this systematic review demonstrated a consistent association between the clinical response as assessed by PASI 75–89 and PASI 90, and disease-specific QoL as assessed by DLQI during a short-term treatment (10–16 weeks); the reduction in PASI scores was directly associated with the improvement in DLQI scores. PASI 90 responders had a greater improvement in DLQI scores as compared with PASI 75–89 responders, which reflected the clinical benefit with QoL. The mean percentage reduction from baseline in DLQI scores was greater in PASI 90 responders as compared with PASI 75–89 responders (91% vs. 75% respectively).[29] This additional 16% improvement in DLQI scores for PASI 90 responders vs. PASI 75–89 responders seems meaningful. Similar and sustained benefit was noted during long-term maintenance phase (26–52 weeks). The 52-week maintenance phase data showed that a significantly greater proportion of PASI 90 responders achieved a DLQI score of 0/1 vs. PASI 75–89 responders,[30] suggesting a sustained improvement in clinical response and QoL.

Meta-analysis estimated a 78% and 90% change from baseline in DLQI scores for PASI 75–89 and PASI 90 responders, respectively, at the primary endpoint of 10–16 weeks. Similarly, a greater proportion of patients achieved a DLQI score of 0/1

© 2016 European Academy of Dermatology and Venereology

14683083, 2017, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jdv.14007 by Western University, Wiley Online Library on [12/06/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

representing no impact on QoL as compared with PASI 75–89 responders (73% vs. 45%). These results are consistent with data reported for individual studies where a better clinical response was associated with an improved QoL.[29,30] Similar results were reported by a recent review where PASI 75 responders had a significant reduction in mean DLQI scores as compared to those having a 50–75% or <50% PASI response.[31]

Overall, a consistent trend of association was observed between improvement in PASI and DLQI scores; a reduction in PASI score was directly related with the improvement in DLQI. This supports the importance from the patients' QoL perspective of achieving the highest possible response, PASI 90. The treatment options provided should include therapies which can achieve higher proportion of patients with PASI 90 or above scores resulting in better QoL for patients with psoriasis.

These analyses provide evidence of superiority in DLQI outcomes in PASI 90 compared to PASI 75–89 responders. However, there are certain limitations that should be considered while interpreting the results of this review. There were limited publications with relevant data corresponding to the spectrum of PASI improvement. Also, the study conclusion was based on studies that reported DLQI data; results of other HRQoL tools were not adequately reported. There was also insufficient evidence available for meta-analysis at the maintenance phase; so, longer term effects are uncertain. There was also lack of real-world evidence to support the conclusions.

It was not possible to perform a meta-analyses looking at correlation between PASI 75-89 and PASI 90 response and mean change DLQI and DLQI score of 0/1 in subgroups stratified by age, sex or other covariates due to the limited number of studies included in the analyses (there were only four studies in the mean change DLQI analysis and three in the DLQI score of 0/1 analysis), and none of the published studies (PSUNRISE, CHAMPION, REVEAL) reported the requisite subgroups. While it would be possible to re-analyse the FIXTURE, ERASURE and CLEAR data to obtain the subgroup information, this would still lead to a very limited evidence base for meta-analysis. Findings showed PASI 90 translated into greater QoL compared with PASI 75–89 in both short- and long-term but the underlying causal explanations remain unknown. More research is needed to explore and interpret the association between PASI and DLQI.

References

1 Baker EL, Coleman CI, Reinhart KM *et al.* Effect of biologic agents on non-PASI outcomes in moderate to severe plaque psoriasis: systematic review and meta-analyses. *Dermatol Ther* 2012; **2**: 9.

2 Wakee M, Nijsten T. Comorbidities in dermatology. *Dermatol Clin* 2009; **27**: 137–147.

3 Wahl A, Loge JH, Wiklund I, Hanestad BR. The burden of psoriasis: a study concerning health-related quality of life among Norwegian adult patients with psoriasis compared with general population norms. *J Am Acad Dermatol* 2000; **43**: 803–808.

4 Gupta MA, Schork NJ, Gupta AK, Kirkby S, Ellis CN. Suicidal ideation in psoriasis. *Int J Dermatol* 1993; **32**: 188–190.

5 Rapp SR, Feldman SR, Exum ML, Fleischer AB Jr, Reboussin DM. Psoriasis causes as much disability as other major medical diseases. *J Am Acad Dermatol* 1999; **41**: 401–407.

6 Sampogna F, Tabolli S, Soderfeldt B, Axtelius B, Aparo U, Abeni D. Measuring quality of life of patients with different clinical types of psoriasis using the SF-36. *Br J Dermatol* 2006; **154**: 844–849.

7 Kurd SK, Troxel AB, Crits-Christoph P, Gelfand JM. The risk of depression, anxiety, and suicidality in patients with psoriasis. *Arch Dermatol* 2010; **146**: 891–895.

8 Khalid JM, Globe G, Fox KM, Chau D, Maguire A, Chiou CF. Treatment and referral patterns for psoriasis in United Kingdom primary care: a retrospective cohort study. *BMC Dermatol* 2013; **13**: 9.

9 Armstrong AW, Schupp C, Wu J, Bebo B. Quality of life and work productivity impairment among psoriasis patients: findings from the National Psoriasis Foundation survey data 2003–2011. *PLoS ONE* 2012; **7**: e52935.

10 Lewis-Beck C, Abouzaid S, Xie L, Baser O, Kim E. Analysis of the relationship between psoriasis symptom severity and quality of life, work productivity, and activity impairment among patients with moderate-to-severe psoriasis using structural equation modeling. *Patient Prefer Adherence* 2013; **7**: 199–205.

11 Gudjonsson JE, Johnston A, Sigmundsdottir H, Valdimarsson H. Immunopathogenic mechanisms in psoriasis. *Clin Exp Immunol* 2014; **135**: 1–8.

12 Nestle FO, Kaplan DH, Barker J. Psoriasis. *N Engl J Med* 2009; **361**: 496–509.

13 Roman M, Madkan VK, Chiu MW. Profile of secukinumab in the treatment of psoriasis: current perspectives. *Ther Clin Risk Manag* 2015; **11**: 1767–1777.

14 Leman J, Burden A. Treatment of severe psoriasis with infliximab. *Ther Clin Risk Manag* 2008; **4**: 1165–1176.

15 Nast A, Jacobs A, Rosumeck S, Werner RN. Efficacy and safety of systemic long-term treatments for moderate-to-severe psoriasis: a systematic review and meta-analysis. *J Invest Dermatol* 2015; **135**: 2641–2648.

16 Puig L. PASI90 response: the new standard in therapeutic efficacy for psoriasis. *J Eur Acad Dermatol Venereol* 2015; **29**: 645–648.

17 Torii H, Sato N, Yoshinari T, Nakagawa H. Japanese infliximab study I. dramatic impact of a psoriasis area and severity index 90 response on the quality of life in patients with psoriasis: an analysis of Japanese clinical trials of infliximab. *J Dermatol* 2012; **39**: 253–259.

18 Finlay AY, Khan GK. Dermatology life quality index (DLQI)-a simple practical measure for routine clinical use. *Clin Exp Dermatol* 1994; **19**: 210–216.

19 Hongbo Y, Thomas CL, Harrison MA, Salek MS, Finlay AY. Translating the science of quality of life into practice: what do dermatology life quality index scores mean? *J Invest Dermatol* 2005; **125**: 659–664.

20 Moher D, Liberati A, Tetzlaff J, Altman DG. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *BMJ* 2009; **339**: b2535.

21 Kalb RE, Blauvelt A, Sofen HL *et al.* Effect of infliximab on health-related quality of life and disease activity by body region in patients with moderate-to-severe psoriasis and inadequate response to etanercept: results from the PSUNRISE trial. *J Drugs Dermatol* 2013; **12**: 874–880.

22 Sutton AJ, Abrams KR, Jones DR, Sheldon TA, Song F. Methods for Meta-Analysis In Medical Research. John Wiley & Sons, London, 2000.

23 Welton NJ, Sutton AJ, Cooper N, Abrams KR, Ades AE. Evidence Synthesis for Decision Making in Healthcare. John Wiley and Sons, London, 2012.

24 Dias S, Welton NJ, Sutton AJ, Ades AE. NICE DSU technical support document 2: A generalised linear modelling framework for pairwise and network meta-analysis of randomised controlled trials. 2011. URL http://www.nicedsu.org.uk. (last accessed: 24 May 2016)

25 Speigelhalter DJ, Best NG, Carlin BP, Van der Linde A. Bayesian measures of model complexity and fit. *JR Stat Soc B* 2002; **64**: 583–639. URL:

© 2016 European Academy of Dermatology and Venereology

14683083, 2017, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jdv.14007 by Western University, Wiley Online Library on [12/06/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

https://classes.soe.ucsc.edu/ams207/Spring10/DIC.pdf (last accessed: 25 May 2016.)

26 Lunn DJ, Jackson CH, Best N, Thomas A, Spiegelhalter D. The BUGS Book: A Practical Introduction to Bayesian Analysis. CRC Press, New York, 2013.

27 Thaçi D, Blauvelt A, Reich K, et al. Secukinumab is superior to ustekinumab in clearing skin of subjects with moderate to severe plaque psoriasis: CLEAR, a randomized controlled trial. *J Am Acad Dermatol.* 2015; **73**: 400–409.

28 Revicki DA, Willian MK, Menter A, Saurat JH, Harnam N, Kaul M. Relationship between clinical response to therapy and health-related quality of life outcomes in patients with moderate to severe plaque psoriasis. *Dermatology* 2008; **216**: 260–270.

29 Navarini AA, Poulin Y, Menter A, Gu Y, Teixeira HD. Analysis of body regions and components of PASI scores during adalimumab or methotrexate treatment for patients with moderate-to-severe psoriasis. *J Drugs Dermatol* 2014; **13**: 554–562.

30 McLeod LD, Mallya UG, Fox T, Zhao Y, Mordin M, Strober B. Psoriasis patients with PASI 90 response achieve greater health-related quality-of-life improvements than those With PASI 75-89 Response. Presented at the European Association of Dermatology and Venereology Congress, Amsterdam, Netherlands (abstr.) 2014.

31 Mattei PL, Corey KC, Kimball AB. Psoriasis area severity index (pasi) and the dermatology life quality index (DLQI): the correlation between disease severity and psychological burden in patients treated with biological therapies. *J Eur Acad Dermatol Venereol* 2014; **28**: 333–337.

© 2016 European Academy of Dermatology and Venereology

14683083, 2017, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jdv.14007 by Western University, Wiley Online Library on [12/06/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License